

Order Flow Imbalance and the Decay of Price Impact in CME Ether Futures

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Abstract

We study Order Flow Imbalance (Cont, Kukanov, and Stoikov, 2014) on CME Ether futures over February 2021 to August 2026. On 28,711 out-of-sample dollar-volume bars, an OLS regression of bar mid-quote change on contemporaneous OFI with Newey-West HAC standard errors yields $\hat{\beta}_0 = +0.215$ ($t = 58.1$, $R^2 = 0.30$); this regression is contemporaneous, so it measures price impact rather than forecasting power. Forward impact is measured from 2,958 signal events, each anchored at the first instant the event is observable from the tape. The signed move is at most +1.7 bp inside ten minutes, grows to +5.5 bp at thirty minutes and roughly +10 bp at four hours (block-bootstrap t between 2.5 and 3.6), and is not identified beyond that horizon; a sign-randomised placebo stays within ± 1.8 bp of zero throughout. The quoted spread at the same instants has a median of 3.5 bp, so one round trip that crosses the spread consumes about a third of the largest predictable move and about the entire move at horizons up to twenty minutes. OFI is informative about price formation on this contract; monetisation is marginal at best.

1 Introduction

Order Flow Imbalance (OFI) aggregates best-bid and best-offer quote arrivals into a signed measure of top-of-book pressure. On liquid US equities, contemporaneous OFI explains roughly two-thirds of bar-level mid-quote variation (Cont, Kukanov, and Stoikov, 2014). The cryptocurrency evidence is thinner: Silantsev (2019) finds trade-flow imbalance dominates short-horizon prices on BitMEX XBT/USD, and Bieganowski and Ślepaczuk (2026) report OFI as the top SHAP feature in CatBoost ensembles on Binance Futures perpetuals. CME’s Ether contract sits apart from both: a regulated venue whose quoted spread, in basis points of price, is an order of magnitude wider than the equity tape’s.

This paper estimates the contemporaneous OFI–price relation on CME ETH futures from February 2021 to August 2026 and then measures forward impact directly at sub-bar to 24-hour horizons. The construction is the level-1 (best bid / best offer) form of Cont, Kukanov, and Stoikov on dollar-volume bars built from TBBO (trade-sampled top-of-book) events (Lopez de Prado, 2018).

2 Data

The panel consists of CME-cleared TBBO data for the front-month ETH futures contract, covering February 2021 through August 2026, with rolls handled by contract stitching.

Bars are constructed in dollar volume, with the threshold calibrated so that a median in-sample trading day produces about ten bars, and that threshold is held fixed when bars are constructed for the out-of-sample period.

*Independent research conducted by the author; the views expressed are the author’s own and not those of Imperial College London.

For each bar the following quantities are computed:

- **ofi**: signed level-1 (best bid / best offer) order flow imbalance, following the construction of Cont, Kukanov, and Stoikov (2014). For each TBBO update i , let p_i^B, p_i^A denote the best-bid and best-ask quotes and q_i^B, q_i^A their sizes. The per-event OFI contribution is $e_i = e_i^B + e_i^A$, where

$$e_i^B = \begin{cases} +q_i^B & \text{if } p_i^B > p_{i-1}^B \\ -q_{i-1}^B & \text{if } p_i^B < p_{i-1}^B \\ q_i^B - q_{i-1}^B & \text{if } p_i^B = p_{i-1}^B \end{cases}, \quad e_i^A = \begin{cases} -q_i^A & \text{if } p_i^A < p_{i-1}^A \\ +q_{i-1}^A & \text{if } p_i^A > p_{i-1}^A \\ -(q_i^A - q_{i-1}^A) & \text{if } p_i^A = p_{i-1}^A. \end{cases}$$

The bar OFI is the unnormalised sum over all events in the bar, $O_t = \sum_{i \in \text{bar } t} e_i$.

- **mid_close**: the mid quote $(p^B + p^A)/2$ at the bar boundary.
- **d_mid**: the change in mid quote since the previous bar.

3 Signal-impact regression

Following Cont, Kukanov, and Stoikov (2014) and Silantyev (2019), bar mid-quote change is regressed on contemporaneous and one-bar lagged OFI:

$$\Delta p_t = \alpha + \beta_0 O_t + \beta_1 O_{t-1} + \varepsilon_t, \quad (1)$$

where $\Delta p_t = m_t - m_{t-1}$ is the change in mid-quote between bar boundaries and O_t is the bar OFI, so β_0 is in dollars of mid-quote change per contract of net imbalance. Standard errors are Newey-West HAC (Newey and West, 1987) with five lags. Table 1 reports the estimates on the full panel, the 2021-02 to 2023-12 in-sample (IS) window, and the 2024-01 to 2026-08 out-of-sample (OOS) window.

Window	n (bars)	$\hat{\beta}_0$	HAC $t(\beta_0)$	$\hat{\beta}_1$	HAC $t(\beta_1)$	R^2
Full sample	41,055	+0.247	+70.2	-0.027	-14.4	0.29
In-sample 2021–23	12,344	+0.300	+44.9	-0.025	-7.0	0.31
Out-of-sample 2024–26	28,711	+0.215	+58.1	-0.026	-13.3	0.30

Table 1: OLS, Newey-West HAC (five lags). $p < 10^{-6}$ on both coefficients in every window.

Both subsamples reject $\beta_0 = 0$ at $p < 10^{-6}$ (Figure 1). The IS and OOS R^2 are nearly identical (0.31 and 0.30), while the slope itself is less stable, moving from +0.300 in-sample to +0.215 out-of-sample. Both R^2 figures are below the two-thirds level of Cont et al. (2014) for liquid US equities.

This regression is contemporaneous: it measures how the mid quote moves together with the flow inside a bar, not whether flow forecasts the next bar. Used on its own as a predictor, lagged OFI explains essentially none of the following bar’s move ($R^2 < 0.001$ out-of-sample), so the content of Table 1 is price impact rather than forecastability. Whether the flow carries any predictive content is a question about horizons inside the bar (mean OOS bar duration ≈ 49 minutes), which the bar regression cannot resolve; Section 3.1 measures it directly.

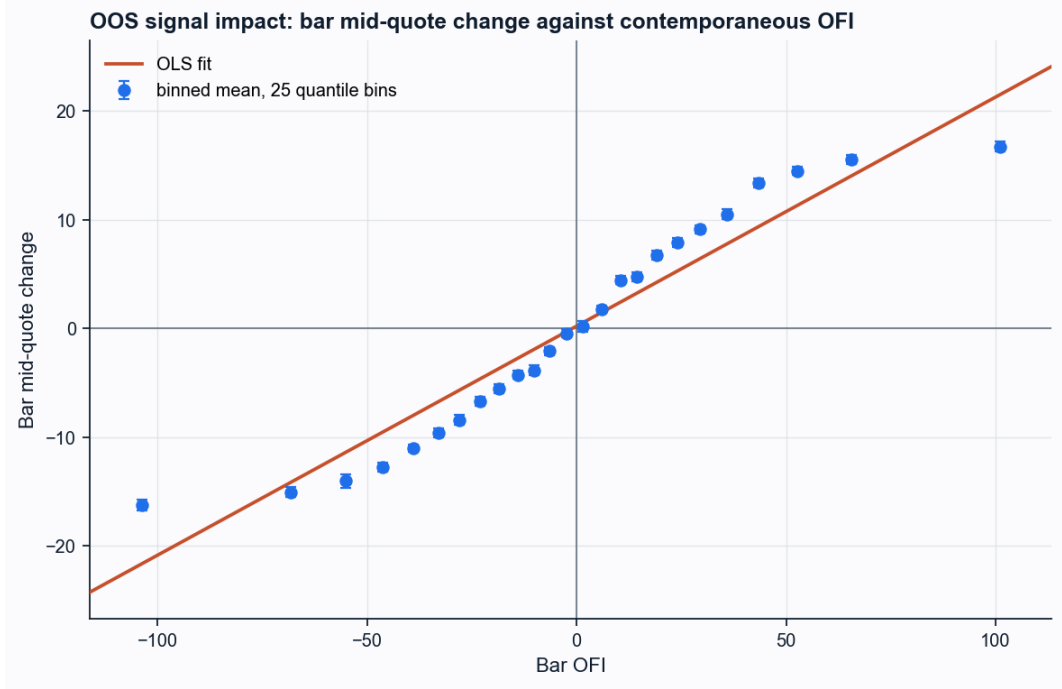


Figure 1: Out-of-sample bar mid-quote change against contemporaneous OFI, 25 quantile bins with ± 1 standard error and the OLS fit. The relationship is monotonic, with impact per unit of flow attenuating at extreme flows — both tail bins fall inside the straight-line fit; full estimates are in Table 1.

3.1 Forward impact after signal events

Forward impact is measured at 21 horizons from 30 seconds to 24 hours after signal events. A signal event is the first bar at which the rolling z -score of cumulative OFI over $K = 5$ bars,

$$z_t = \frac{C_t - \mu_t}{\sigma_t}, \quad C_t = \sum_{s=t-K+1}^t O_s, \quad (2)$$

crosses $|z_t| \geq 1$ from below, with μ_t and σ_t the trailing mean and standard deviation of C over a 200-bar window shifted by one bar. The sign of z_t gives the event direction; there are 2,958 out-of-sample events. This defines a timestamp and a direction only — no position or holding rule is involved. These parameters were originally selected for a trading application of the same signal, so the profile is checked against that choice: varying $K \in \{3, 5, 10\}$ at threshold 1, and the threshold across $\{1, 1.5, 2\}$ at $K = 5$ (re-measured at minute resolution for comparability across configurations), the four-hour mean ranges from +8.7 to +11.8 bp while the one-minute means stay near zero throughout.

Timing is handled conservatively. Bars are aggregated from one-minute buckets, and a bar is complete only once the contract’s cumulative dollar volume crosses the bar threshold, which happens at a specific trade after the bar’s final minute; $t = 0$ is anchored at that trade, the first instant the event is observable, and the mid quote and spread are taken there. For each horizon the mid is the last quote at or before the target time, discarded if more than 120 seconds stale (e.g. past the daily 5–6 pm ET session break), so the usable sample varies by horizon and the attrition is concentrated at particular times of day. On the 483 events measurable at every horizon, the point estimates trend upward with horizon and show no detectable reversal, so behaviour beyond roughly four hours is not identified by this design and no claim is made about it.

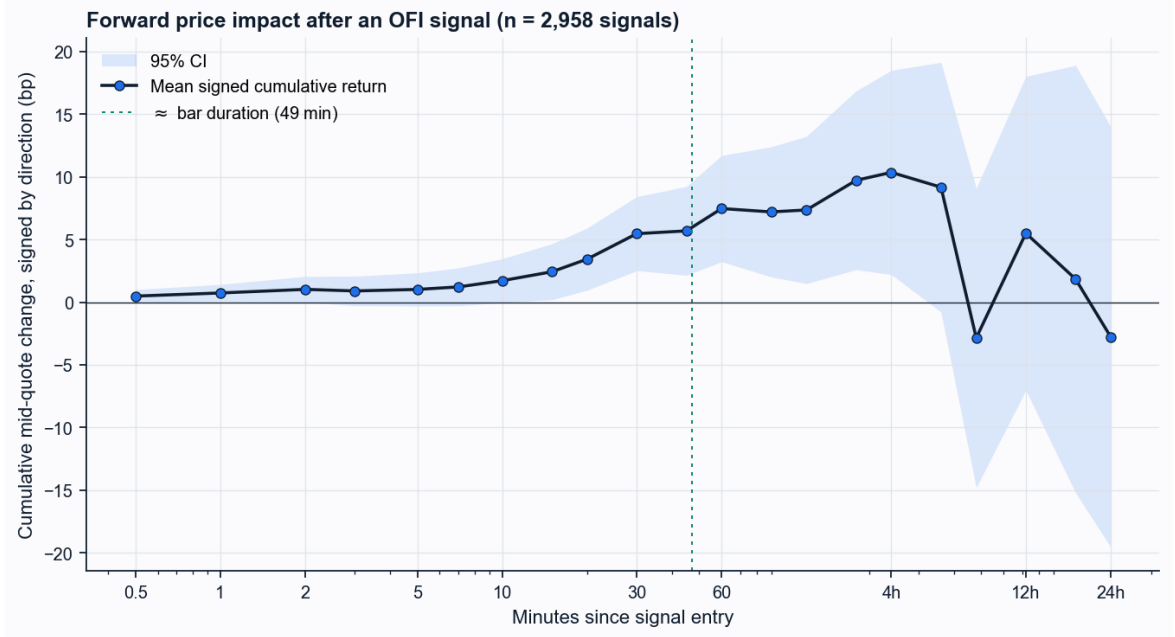


Figure 2: Signed forward return after a signal event, OOS, with the block-bootstrap 95% band. The move stays below +2 bp inside ten minutes, grows through the 30-minute to 4-hour range, and is unidentified beyond it.

Horizon	Signed return (bp)	Block-bootstrap t	95% CI (bp)	n
0.5 min	+0.49	1.9	$[-0.0, +1.0]$	2,958
1 min	+0.74	2.1	$[+0.1, +1.4]$	2,958
2 min	+1.03	2.0	$[+0.0, +2.1]$	2,958
5 min	+1.02	1.5	$[-0.2, +2.4]$	2,850
10 min	+1.72	1.9	$[-0.0, +3.5]$	2,807
20 min	+3.44	2.7	$[+1.0, +5.9]$	2,776
30 min	+5.48	3.6	$[+2.6, +8.6]$	2,758
1 hour	+7.48	3.5	$[+3.6, +11.8]$	2,684
2 hours	+7.37	2.5	$[+1.9, +13.4]$	2,475
3 hours	+9.73	2.7	$[+2.6, +17.0]$	2,434
4 hours	+10.37	2.5	$[+2.5, +18.5]$	2,410
6 hours	+9.18	1.8	$[-0.3, +19.7]$	2,230
8 hours	-2.85	-0.5	$[-14.9, +8.8]$	2,030
12 hours	+5.51	0.9	$[-7.0, +17.7]$	2,109
24 hours	-2.77	-0.3	$[-19.8, +14.1]$	2,149

Table 2: Mean forward signed return at fixed horizons after an OOS signal event (15 of the 21 measured horizons shown; the omitted rows at 3, 7, 15, 45, 90 minutes and 18 hours lie on the same path, and none exceeds the four-hour figure). The t -statistic and 95% interval are from a by-day block bootstrap (2,000 resamples), which keeps each day’s overlapping forward windows together. A non-overlapping subsample (events at least one horizon apart) gives $t = 2.5$ at four hours.

Figure 2 plots the profile and Table 2 reports it. A sign-randomised placebo at the same timestamps, averaged over 50 seeds, stays within ± 1.8 bp of zero at every horizon and its 95% band always covers zero, so the curve is not market drift. Standard errors and confidence intervals are from a block bootstrap over trading days: the forward windows of nearby events overlap, so a cross-sectional standard error would understate uncertainty at long horizons. A non-overlapping subsample, keeping events at least one horizon apart, is the cross-check.

The shape is the result. Inside ten minutes the signed move is at most +1.7 bp, half the median quoted spread at the event instants. It grows through the half-hour (+5.5 bp, $t = 3.6$) and

the hour (+7.5 bp, $t = 3.5$) and reaches roughly +10 bp around four hours ($t = 2.5$, 95% CI [+2.5, +18.5]). The four-hour value is the largest among the significant horizons and is read as the scale of the move, not as a tested peak location: it is one of 21 correlated estimates, and beyond it the pooled estimates are noisy while the fixed-horizon subsample shows no reversal.

4 Economic magnitude

The quoted spread, measured at the 2,958 event instants themselves, has a median of 3.5 bp with an interquartile range of 2.4 to 5.3 bp. A round trip that crosses the spread therefore costs about one median spread — roughly a third of the four-hour move, while the twenty-minute move (+3.4 bp) is about equal to it and everything shorter is smaller. Capturing the hours-scale drift means holding through a window whose 95% interval spans +2.5 to +18.5 bp, before any slippage or latency.

Posting passively rather than crossing avoids the spread but introduces adverse selection: a resting order fills when the market comes to it, which on a flow-following signal is when the flow has already reversed. Monetisation is therefore marginal at best: the move inside twenty minutes is at best about equal to the spread, the half-hour to two-hour moves clear it by less than a factor of 2.2, and even the four-hour move exceeds it only about threefold.

5 Conclusion

The contemporaneous OFI–price relation is strong on CME Ether futures (HAC $t = 58.1$ out-of-sample), and it is a statement about price impact, not forecastability. The forward measurement gives the predictive content its honest size: a sub-spread move inside ten minutes, then a drift that builds to roughly +10 bp over four hours, robust to the event-definition parameters and flat under a sign-randomised placebo, and unidentified beyond that horizon. Set against a median quoted spread of 3.5 bp at the same instants, the relationship is informative about price formation while offering at most a marginal margin over trading costs.

The tape bounds what can be measured. TBBO is trade-sampled, so the book is observed only when a trade prints — the median staleness of the matched quote at measurement horizons is nine seconds — and it carries no depth beyond the touch, so execution cannot be treated properly here. A quote-level feed would sharpen the sub-minute horizons and support a proper execution study, which is the natural extension.

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